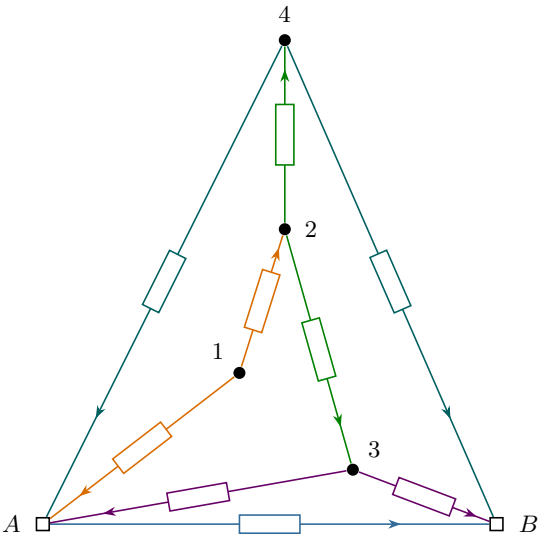
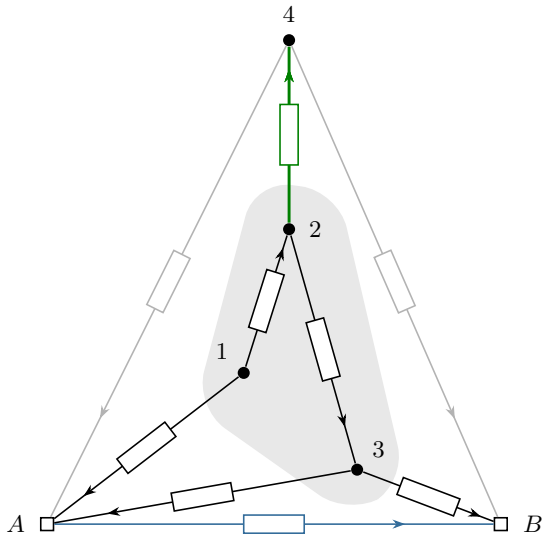
(a) Two resistors for each interaction vertex



$1\colon 12, 1A \quad 2\colon 23, 24 \quad 3\colon 3A, 3B \quad 4\colon 4A, 4B$

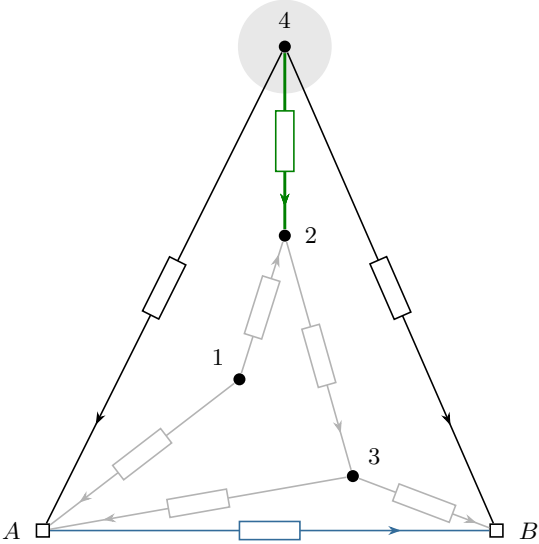
(b) Cutting $24 = (2, 4)$: $S = \{1, 2, 3\}$



$$\underbrace{2|S|}_{6} - \underbrace{(|S| - 1)}_{12,\,23} - \underbrace{|\Gamma_1(S, \{A, B\})|}_{1A,\,3A,\,3B} = 1$$

deficit 1: $24 \mapsto e(0) = 2$

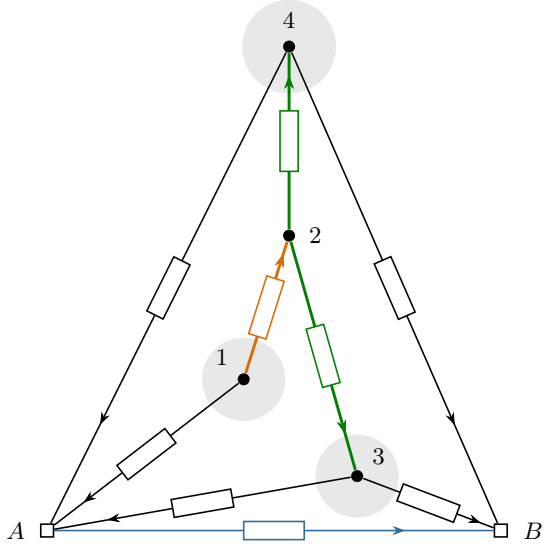
(c) Cutting $24 = (4, 2)$: $S = \{4\}$



$$\underbrace{2|S|}_{2} - \underbrace{(|S| - 1)}_{0} - \underbrace{|\Gamma_1(S, \{A, B\})|}_{4A,\,4B} = 0$$

deficit 0: $24 \mapsto e(1) = 2$

(d) Removing vertex 2



deficits: $\{1\}\colon 1, \quad \{3\}\colon 0, \quad \{4\}\colon 0$

$12 \mapsto 1, \quad 23, 24 \mapsto 2$: two resistors at 2